# TQM (19MEE347) – Two Predicted End-Semester Papers (100 Marks Each)

Amrita School of Engineering, Chennai · Built from your CDP, assignments, quiz, teacher hints, Unit 1–3 slides and priority tables

> **Important:** these are **predictions (estimates), not guaranteed or leaked questions.** No file you uploaded states the official question-paper pattern, so the layout below is a stated working assumption.

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# 1. Document analysis and syllabus coverage

## 1.1 Files scanned and what each one gave

| File(s) | Content | Status |
|---|---|---|
| 19MEE347_TQM_CDP_V1.pdf | Course Delivery Plan: objective, 4 COs, syllabus, 46 lectures, evaluation scheme | Readable |
| Assignment_1.pdf | Q1 quality dimensions + quality costs in quality planning; Q2 TQM implementation (leadership, council, statements, strategic planning, Deming, barriers) | Readable |
| Assignment_2.pdf | Taguchi loss-function problems, numbered Q1–Q4 and Q9 | Readable, **Q5–Q8 are missing from the PDF** |
| Quiz_Question.png | Set A (three spheres of quality; X-bar/R chart) and Set B (Deming; p-chart) | Readable (photo) |
| EndSem_info.jpg, Unit_3_info.png | Teacher hints: Unit 3 = 20–25 marks, numericals = 25–30 marks; study FMEA and practise its numericals | Readable |

| File(s) | Content | Status |
|---|---|---|
| Screenshot_2026-10-23_113601.jpg, 1792749869636_image.png | Your Unit 1 and Unit 2 priority tables | Readable |
| Unit_1.pdf (87 slides), Unit_2_Large.pdf (139 slides) | Unit 1 and Unit 2 teaching slides | Readable |
| 8 Unit 3 PPTs (QFD 30, QLF 19, TPM 29, ISO 9000 20, Audit 21, ISO 14000 12, 5S 18, FMEA 30 slides) | Unit 3 teaching slides | Readable (TPM is mostly images; I read them visually) |
| TQM_Exam_Marks_Split.md | An earlier ChatGPT chat with planning estimates | Readable; **estimates only, not official** |

**Not provided / gaps:** the official paper pattern (sections, choices, marks per question); any previous-year paper; Assignment 2 Q5–Q8; the "Unit 3 notes.rar" (only the 8 PPTs reached me); slides for QS 9000. Unit 2 slides do **not** cover process capability, statistical fundamentals, customer retention, employee involvement, performance appraisal, supplier partnership or performance measures, so I based those on the syllabus line and standard TQM content.

## 1.2 Evidence on weightage

| Evidence | What it says |
|---|---|
| CDP syllabus | Unit 1: definition/dimensions, quality planning/costs, TQM principles, leadership, council, statements, strategic planning, Deming, barriers. Unit 2: customer satisfaction/retention, employee involvement, appraisal, CPI, supplier partnership, performance measures, seven tools, statistics, control charts, process capability, Six Sigma, new seven tools, benchmarking. Unit 3: QFD, Taguchi, TPM, FMEA, need for quality systems, ISO 9000:2000, implementation, documentation, auditing, QS 9000, ISO 14000 |
| CDP lecture count | Unit 1 = 11 lectures (24%), Unit 2 = 19 (41%), Unit 3 = 16 (35%) of 46 |

| Evidence | What it says |
|---|---|
| CDP outcomes | CO1 principles (lectures 1–8); CO2 performance measures with tools (lectures 9–30); CO3 QFD, Taguchi, TPM, FMEA (31–38); CO4 quality systems (39–45) |
| Teacher hints | Unit 3 = 20–25 marks; numericals = 25–30 marks; FMEA to be studied with numericals |
| Assignments and quiz | Deming and control charts repeat; Taguchi is the whole of Assignment 2; Assignment 1 targets dimensions, costs and TQM implementation |
| CDP evaluation | End-semester is shown as 30 of the 100 course marks; you describe a 100-mark paper, so the paper is probably scaled |

**Weighting I used (estimate):** Unit 1 ≈ 21, Unit 2 ≈ 55, Unit 3 ≈ 24, with 28–30 numerical marks inside those totals. Unit 3 respects your teacher's 20–25 hint even though the lecture count alone would suggest more; Unit 2 gets the most marks because it holds most numerical topics.

## 1.3 Paper pattern used (assumption, not official)

| Part | Questions | Marks each | Total | Choice |
|---|---|---|---|---|
| A | Q1–Q10, short answer | 2 | 20 | All compulsory |
| B | Q11–Q16, medium answer | 5 | 30 | All compulsory |
| C | Q17–Q21, long answer | 10 | 50 | Answer **either (a) or (b)** of each question |
| **Total** | | | **100** | |

Internal choices in Part C are built so both options are from the **same unit and the same type** (both numerical or both theory), so the marks and numerical share do not change with the student's choice.

Tags used below: **[U1/U2/U3]** unit; **[N]** numerical; 🔴 very important, 🟡 moderately important, 🟢 lower priority (from your priority tables).

# 1.4 Coverage matrix (where each priority topic appears)

| Topic | Priority | Set 1 | Set 2 |
|---|---|---|---|
| Deming philosophy / 14 points | 🔴 | Q17(a) | Q11 |
| Quality dimensions and quality costs | 🔴 | Q3, Q17(b) | Q3 (numerical) |
| Leadership, council, strategic planning, statements, barriers | 🔴 / 🟡 | Q2, Q11 | Q1, Q17(a) |
| Quality planning, TQM history and principles | 🟡 | Q17(b) | Q17(b) |
| Three spheres of quality | 🟡 | – | Q2 |
| Seven tools of quality | 🔴 | Q5, Q19(a) | Q5, Q13 |
| Statistical fundamentals | 🔴 | Q7 | Q7 |
| X-bar/R charts | 🔴 | Q18(a) | Q18(a) |
| p, np, c charts | 🔴 | Q18(b) p | Q14 c, Q18(b) np |
| Process capability Cp/Cpk | 🔴 | Q14 | Q8, Q19(b) |
| Six Sigma and DPMO | 🟡 | Q8, Q19(b) | Q12 |
| Customer satisfaction / retention | 🟡 | Q4, Q12 | Q4, Q20(a) |
| CPI / PDCA | 🟡 | Q13 | – |
| Benchmarking | 🟡 | Q20(a) | Q6 |
| New seven management tools | 🟡 | Q6, Q20(b) | – |
| Employee involvement, appraisal, supplier partnership, performance measures | 🟢 | – | Q20(b) |
| FMEA | 🔴 | Q10, Q21(a) | Q15 |
| Taguchi loss function | 🔴 | Q15 | Q10 |
| QFD / House of Quality | 🔴 | Q9 | Q21(a) |
| TPM / OEE | 🔴 | Q21(b) | Q21(b) |
| ISO 9000:2000 | 🟡 | Q16 | – |
| Quality audit | 🟡 | – | Q9, Q16 |

| Topic | Priority | Set 1 | Set 2 |
|---|---|---|---|
| ISO 14000 | 🟡 | – | Q9 |

**Not placed in either set (reserve topics, so still prepare them):** 5S, QS 9000, Taguchi robust-design theory, quality gurus, and the Unit 3 items "need for quality systems / documentation / implementation" beyond what ISO 9000 covers. Reason: only five Unit 3 slots exist per paper and these have the lowest estimated likelihood (5S is not in the syllabus text; QS 9000 has no slides).

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# 2. SET 1 – Predicted Question Paper (100 Marks)

**Course:** 19MEE347 Total Quality Management · **Maximum marks:** 100 **Instructions:** Answer all questions in Parts A and B. In Part C answer **either (a) or (b)** of each question. Assume suitable data if needed.

## Part A – Short answer (10 × 2 = 20 marks)

1. [U1] Define quality, and give the definitions of quality by any two quality gurus. (2) 🟢
2. [U1] List any four barriers to TQM implementation. (2) 🔴
3. [U1] Name the four categories of quality costs and give one example for each. (2) 🔴
4. [U2] State the three categories of customer requirements in Kano's model. (2) 🟡
5. [U2] State the Pareto principle and the purpose of a Pareto chart. (2) 🔴
6. [U2] List any four of the new seven management tools. (2) 🟡
7. [U2] Distinguish between common causes and special causes of variation. (2) 🔴
8. [U2][N] A batch has 500 units, each with 4 opportunities for a defect, and 15 defects are found. Calculate the DPMO. (2) 🟡
9. [U3] State the four phases of Quality Function Deployment. (2) 🔴
10. [U3][N] In an FMEA, a failure mode has severity 7, occurrence 4 and detection 6. Calculate its RPN. After a corrective action detection improves to 2; find the revised RPN. (2) 🔴

## Part B – Medium answer (6 × 5 = 30 marks)

11. [U1] Explain the composition and duties of the quality council. (5) 🔴
12. [U2] Explain the factors that influence customer perception of quality and the tools used to collect customer feedback. (5) 🟡

13. [U2] Explain the PDSA cycle and its role in continuous process improvement. (5) 🟡
14. [U2][N] A shaft has specification limits of 19.0 mm and 25.0 mm. A stable process has a mean of 21.5 mm and a standard deviation of 0.9 mm. Calculate Cp, CPU, CPL and Cpk and comment on the capability of the process. (Cp 1; CPU and CPL 2; Cpk 1; comment 1) (5) 🔴
15. [U3][N] A part has a target of 50 mm. The customer loses ₹16 when the part is 0.4 mm away from the target. (i) Find the loss coefficient k. (1) (ii) Find the loss for a part measuring 50.3 mm. (1) (iii) Find the average loss per part for a process with mean 50.2 mm and standard deviation 0.1 mm. (2) (iv) A factory adjustment costs ₹1; beyond what deviation from target is it economical to adjust? (1) (5) 🔴
16. [U3] Explain the five sections of ISO 9001:2000 and the levels of documentation. (5) 🟡

## Part C – Long answer (5 × 10 = 50 marks; answer either (a) or (b))

17. [U1] 🔴 (a) Explain Deming's philosophy of quality, including the chain reaction, and discuss any five of his 14 points. (10) **OR** (b) Analyse the role of quality dimensions (product and service) and the four quality costs in an effective quality planning framework, and show how the costs influence organisational performance. (10)
18. [U2][N] 🔴 (a) Ten samples of size 5 gave the following means and ranges. Using A2 = 0.577, D3 = 0 and D4 = 2.114, calculate the centre lines and control limits for the X-bar and R charts and comment on the state of control. (X-double-bar and R-bar 2; X-bar limits 3; R limits 3; conclusion 2) (10)

| Sample | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Mean | 52 | 50 | 49 | 53 | 51 | 48 | 54 | 50 | 47 | 56 |
| Range | 6 | 5 | 7 | 4 | 6 | 8 | 5 | 6 | 7 | 5 |

**OR** (b) Twelve samples of 100 units each had the following numbers of defectives: 5, 7, 4, 6, 8, 3, 6, 5, 7, 16, 5, 4. Determine the centre line and control limits of the p-chart, plot or compare the points and state whether the process is in statistical control. (p-bar 2; limits 4; comparison 2; conclusion 2) (10)

19. [U2] 🔴 / 🟡 (a) Explain the seven tools of quality, with one application for each. (10) 🔴 **OR** (b) Explain Six Sigma: concept (3.4 DPMO), DMAIC and DMADV methods, belt hierarchy and benefits. (10) 🟡
20. [U2] 🟡 (a) Define benchmarking and explain its types, process steps and benefits. (10) **OR** (b) Explain the new seven management tools and state when each is used. (10)

21. [U3] 🔴 (each option = 4 marks numerical + 6 marks theory) (a) FMEA: (i) For the failure modes below, calculate the RPN of each and rank them. (4) (ii) Explain the FMEA procedure and differentiate design FMEA from process FMEA. (6)

| Failure mode | Severity | Occurrence | Detection |
|---|---|---|---|
| Weld crack | 9 | 3 | 6 |
| Paint blemish | 3 | 7 | 2 |
| Wrong bolt torque | 7 | 4 | 7 |

**OR** (b) TPM: (i) Planned production time is 480 min, downtime 60 min, ideal cycle time 0.5 min/unit, units produced 700 and good units 665. Calculate availability, performance, quality and OEE. (4) (ii) Explain the eight pillars of TPM and the six major losses. (6)

**Set 1 mark check:** Part A 20 + Part B 30 + Part C 50 = **100**. Units: U1 = 6 + 5 + 10 = **21**; U2 = 10 + 15 + 30 = **55**; U3 = 4 + 10 + 10 = **24**. Numerical: Q8 (2) + Q10 (2) + Q14 (5) + Q15 (5) + Q18 (10) + Q21 (4) = **28**.

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# 3. SET 2 – Predicted Question Paper (100 Marks)

**Course:** 19MEE347 Total Quality Management · **Maximum marks:** 100 **Instructions:** Answer all questions in Parts A and B. In Part C answer **either (a) or (b)** of each question. Assume suitable data if needed.

## Part A – Short answer (10 × 2 = 20 marks)

1. [U1] Differentiate between vision, mission and quality policy. (2) 🟡
2. [U1] Name the three spheres of quality and give one activity under each. (2) 🟡
3. [U1][N] A company spends ₹1.5 lakh on prevention, ₹2.5 lakh on appraisal, ₹3.0 lakh on internal failure and ₹5.0 lakh on external failure. Sales are ₹250 lakh. Find the total cost of quality and express it as a percentage of sales. (2) 🔴
4. [U2] Differentiate between internal and external customers with one example each. (2) 🟡
5. [U2] State the purpose of a histogram and of a scatter diagram. (2) 🔴
6. [U2] Define benchmarking and name any two types of benchmarking. (2) 🟡
7. [U2] For a normal distribution, state the percentage of data within ±1σ, ±2σ and ±3σ, and why 3σ control limits are used. (2) 🔴

8. [U2][N] A process has USL = 50 mm, LSL = 40 mm and σ = 1.2 mm. Calculate Cp and comment. (2) 🔴
9. [U3] Define quality audit and name its three types by the party performing it. (2) 🟡
10. [U3][N] The target length of a part is 200 mm and the loss is ₹40 when the length deviates by 4 mm. Find k and the loss for a part of length 197 mm. (2) 🔴

## Part B – Medium answer (6 × 5 = 30 marks)

11. [U1] Explain Deming's chain reaction and any three of his 14 points. (5) 🔴
12. [U2] Explain the DMAIC and DMADV methods of Six Sigma. (5) 🟡
13. [U2] Explain the cause-and-effect (fishbone) diagram with the 6M categories and a suitable example. (5) 🔴
14. [U2][N] The number of defects in 12 consecutive bolts of fabric was 6, 9, 4, 7, 5, 8, 12, 6, 3, 7, 5, 8. Construct the c-chart limits and state whether the process is in control. (c-bar 1; limits 2; comparison 1; conclusion 1) (5) 🔴
15. [U3][N] Four failure modes were rated as shown. (i) Calculate the RPN of each. (2) (ii) Rank them. (1) (iii) Which mode must be acted on irrespective of its RPN, and why? (1) (iv) For the mode with the highest RPN, a new check improves detection from 6 to 3; find its revised RPN. (1) (5) 🔴

| Failure mode | Severity | Occurrence | Detection |
|---|---|---|---|
| A: Hydraulic hose leak | 10 | 2 | 5 |
| B: Sensor drift | 6 | 5 | 6 |
| C: Label misprint | 3 | 6 | 2 |
| D: Connector loose | 8 | 2 | 8 |

16. [U3] Explain the types of quality audit (by party and by sub-type) and the steps in a quality audit. (5) 🟡

## Part C – Long answer (5 × 10 = 50 marks; answer either (a) or (b))

17. [U1] (a) Analyse the factors affecting successful TQM implementation, discussing the roles of leadership, quality council, quality statements and strategic planning, and the major barriers. (10) 🔴 **OR** (b) Trace the historical evolution of quality management and explain the eight quality management principles. (10) 🟡

18. [U2][N] 🔴 (a) Twelve samples of size 4 gave the following means and ranges. Using A2 = 0.729, D3 = 0 and D4 = 2.282, calculate the control limits for the X-bar and R charts and comment on the process. (centre lines 2; X-bar limits 3; R limits 3; conclusion 2) (10)

| Sample | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mean | 12.5 | 12.1 | 12.8 | 13.0 | 12.4 | 12.2 | 12.9 | 12.6 | 12.3 | 13.4 | 12.0 | 12.7 |
| Range | 0.8 | 0.6 | 1.0 | 0.7 | 0.9 | 0.5 | 0.8 | 0.6 | 0.7 | 1.1 | 0.4 | 0.9 |

**OR** (b) Twelve samples of 200 units each had the following numbers of defectives: 8, 6, 9, 5, 7, 10, 4, 8, 6, 9, 7, 20. Construct the np-chart (centre line and limits) and state whether the process is in statistical control. (np-bar 2; limits 4; comparison 2; conclusion 2) (10)

19. [U2] 🔴 (a) Explain the types of control charts for variables and attributes, the logic for selecting a chart, and the tests for an out-of-control process. (10) **OR** (b) Explain process capability: the meaning of Cp and Cpk, their formulas and interpretation, and the difference between them. (10)

20. [U2] 🟡 / 🟢 (a) Explain customer satisfaction and retention: Teboul and Kano models, customer perception, complaints and feedback tools, and ways to improve retention. (10) 🟡 **OR** (b) Explain employee involvement, performance appraisal, supplier partnership and performance measures in TQM. (10) 🟢

21. [U3] 🔴 (each option = 4 marks numerical + 6 marks theory) (a) QFD: (i) Customer importance weights are 5, 4 and 2. Using the relationship matrix below (9 strong, 3 medium, 1 weak), calculate the technical importance of each HOW and its relative percentage, and state which HOW has top priority. (4) (ii) Explain the House of Quality with a neat sketch and state the benefits of QFD. (6)

| Customer requirement (importance) | HOW 1 | HOW 2 | HOW 3 |
|---|---|---|---|
| Easy to use (5) | 9 | 1 | 3 |
| Durable (4) | 3 | 9 | 1 |
| Low price (2) | 1 | 3 | 9 |

**OR** (b) TPM: (i) Planned time is 420 min, downtime 42 min, ideal cycle time 0.6 min/unit, units produced 600 and good units 570. Calculate availability, performance, quality and OEE. (4) (ii) Explain the goals and strategies of TPM, the three zeros, and the difference between TPM and TQM. (6)

**Set 2 mark check:** Part A 20 + Part B 30 + Part C 50 = **100**. Units: U1 = 6 + 5 + 10 = **21**; U2 = 10 + 15 + 30 = **55**; U3 = 4 + 10 + 10 = **24**. Numerical: Q3 (2) + Q8 (2) + Q10 (2) + Q14 (5) + Q15 (5) + Q18 (10) + Q21 (4) = **30**.

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# 4. Verification summary

| Check | Set 1 | Set 2 |
|---|---|---|
| Total marks (answered) | 100 | 100 |
| Part A / B / C | 20 / 30 / 50 | 20 / 30 / 50 |
| Unit 1 / 2 / 3 marks | 21 / 55 / 24 | 21 / 55 / 24 |
| Numerical marks (target 25–30) | 28 | 30 |
| Unit 3 within teacher's 20–25 hint | 24 ✓ | 24 ✓ |
| Part C choices same unit and type | ✓ | ✓ |
| Different data and wording from the other set | ✓ | ✓ |

Numerical marks are counted inside the unit marks, not added to them. Topics such as Deming, X-bar/R and Taguchi appear in both sets on purpose (they are the most repeated in your evidence) but with different question forms and data.

## Numerical answer key (for self-checking; verified by recalculation)

| Question | Answer |
|---|---|
| S1 Q8 | DPMO = 15 ÷ (500×4) × $10^6$ = **7,500** (about 3.9 sigma) |
| S1 Q10 | RPN = 7×4×6 = **168**; revised = 7×4×2 = **56** |
| S1 Q14 | Cp = 6 ÷ 5.4 = **1.11**; CPU = **1.30**; CPL = **0.93**; Cpk = **0.93** (below 1, not capable; off-centre toward LSL) |
| S1 Q15 | k = 16 ÷ $0.4^2$ = **100**; L(50.3) = **₹9**; average loss = $100(0.1^2 + 0.2^2)$ = **₹5**; break-even = √(1/100) = **0.1 mm** |

| Question | Answer |
|---|---|
| S1 Q18(a) | X-double-bar = 51.0, R-bar = 5.9; X-bar UCL **54.404**, LCL **47.596**; R UCL **12.473**, LCL **0**. Means 47 and 56 are outside, so X-bar chart out of control; R chart in control |
| S1 Q18(b) | p-bar = 76 ÷ 1200 = **0.0633**; UCL **0.1364**; LCL = −0.0097 → **0**. Sample 10 (0.16) is outside, so out of control |
| S1 Q21(a) | RPN: weld crack **162**; paint blemish **42**; wrong torque **196**. Rank: wrong torque, weld crack, paint blemish |
| S1 Q21(b) | Availability **87.5%**; performance **83.3%**; quality **95%**; OEE **69.3%** |
| S2 Q3 | Total COQ = **₹12 lakh** = **4.8%** of sales (cost of poor quality ₹8 lakh) |
| S2 Q7 | 68.26%, 95.46%, 99.73% |
| S2 Q8 | Cp = 10 ÷ 7.2 = **1.39** (potentially capable if centred) |
| S2 Q10 | k = 40 ÷ 16 = **2.5**; L(197) = 2.5 × 9 = **₹22.5** |
| S2 Q14 | c-bar = 80 ÷ 12 = **6.67**; UCL **14.41**; LCL = −1.08 → **0**; all points inside, so in control |
| S2 Q15 | RPN: A **100**, B **180**, C **36**, D **128**. Rank: B, D, A, C. A has severity 10 so it must be addressed anyway. Revised B = 6×5×3 = **90** |
| S2 Q18(a) | X-double-bar = 12.575, R-bar = 0.75; X-bar UCL **13.122**, LCL **12.028**; R UCL **1.712**, LCL **0**. Means 13.4 and 12.0 are outside, so X-bar chart out of control; R chart in control |
| S2 Q18(b) | np-bar = 99 ÷ 12 = **8.25**; σ = √(8.25×0.959) = 2.81; UCL **16.69**; LCL = −0.19 → **0**. Sample 12 (20) is outside, so out of control |
| S2 Q21(a) | HOW 1 = 45+12+2 = **59**; HOW 2 = 5+36+6 = **47**; HOW 3 = 15+4+18 = **37**; total 143 → **41.3%, 32.9%, 25.9%**; HOW 1 is top priority |
| S2 Q21(b) | Availability **90%**; performance **95.2%**; quality **95%**; OEE **81.4%** |

# 5. Most important topics to prepare first (estimates)

1. **X-bar/R charts, then p, np and c charts** – numerical, in both quiz sets, about 15–20 marks in total. 🔴
2. **Taguchi loss function** – all of Assignment 2; practise k, loss, average loss and break-even adjustment. 🔴
3. **FMEA** – named by your teacher; learn the 7 steps, the 1–10 scales and RPN problems, including the severity-10 rule. 🔴
4. **Deming's philosophy and 14 points** – repeated in Assignment 1 and quiz Set B. 🔴
5. **Quality dimensions and the four quality costs** – Assignment 1 Q1, with a small cost calculation possible. 🔴
6. **Process capability (Cp, Cpk)** and **statistical fundamentals (normal curve, common vs special causes)** – the base for the numericals. 🔴
7. **Seven tools of quality** – plus one Pareto and one fishbone example. 🔴
8. **QFD House of Quality and TPM (8 pillars, 6 losses, OEE)** – both have numerical hooks. 🔴
9. **Leadership, quality council, strategic planning, barriers** – Assignment 1 Q2. 🔴
10. **Moderate topics next:** Six Sigma/DPMO, benchmarking, new seven tools, customer satisfaction/Kano, PDCA, ISO 9000:2000, quality audit, ISO 14000. 🟡
11. **Skim last:** employee involvement, appraisal, supplier partnership, performance measures, 5S, QS 9000. 🟢

> **Reminder:** the two papers, the paper pattern, the weightages and all likelihood tags are my estimates from your files. The only teacher-sourced facts are Unit 3 = 20–25 marks, numericals = 25–30 marks, and the FMEA message.